Test report 17 – Stencil font "cad stencil mono"

Plan 17.7, step 3b: custom monospaced stencil font as the default font of `text()` · cad-font (`stencil.rs`), cad-eval (`text()` parameters), tests in cad-font, cad-eval, cad-engine, compat-tests, Playwright; fix for Playwright C16/C20 · 2026-10-24 · branch worktree-agent-a84f54f259309a87b · commit 436b868 · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores

Glyphs without holes	cargo test --workspace	clippy · fmt	Vitest · Playwright	Sign, kernel native	Wasm cad_wasm
125 / 125	**529 / 529**	**clean · clean**	**221 · 106 / 106**	**1,791 ms**	**+41.5 KiB**
Regular and Bold; 8 parameter combinations in the test	13 of them new/changed	-D warnings	new: K01–K03; C16 fixed	Allerta Stencil 3,419 ms	gzip +15.0 KiB

1 Task and result

Plan 17.4 calls for a custom, very simple monospaced stencil font made of lines and arcs as the default font, with bridges so that no islands fall out. Implemented:

- **"cad stencil mono"** (Regular, Bold) in `crates/cad-font/src/stencil.rs`: 125 glyphs – the whole required character set `cad_font::REQUIRED_CHARS` without fallback, plus `’ ” $`. Each glyph is a sequence of center lines (lines, tangential circular arcs), drawn with a pen of the stroke width (flat ends, miter), plus dots, minus bridges, trimmed at the baseline and cap line. The contour is computed exactly by `cad_kernel::region2` (pieces of the lines and arcs); STEP and the kernel see lines and circles.
- **No holes:** every glyph, rendered to a profile, has 0 holes – the counters of O, A, B, D, P, R, 0, 4, 6, 8, 9, Ä, ö, @, %, &, §, °, Ø ... are connected to the outside via bridges. Checked in cad-font for 8 combinations of stroke width and bridge width (regions without clockwise loops, closed loops, no slivers, no overlaps) and in the engine for every glyph via `cad_engine::region_2d`.
- **Parameters** `text(…, stroke_width = …, bridge_width = …)` in mm: stroke width default 0.14·`size` (Bold 0.2·`size`), at most 0.2·`size`; bridge width default 0.1·`size`, 0 (no bridges – closed counters like a normal font) or 0.03...0.2·`size`. Values outside: warning at the argument, clamped. With other fonts: warning "wirkt nur mit ..." ("only takes effect with ...").
- **Metrics** as for the other fonts: em = `size`/0.72, cap height = `size` (720/1000 em), advance 600/1000 em × 1000/1024 – exactly as JetBrains Mono, so texts stay as wide as with the previous default font.
- **Default font:** `text()` without `font` and with an unknown family uses the stencil font (previously JetBrains Mono); missing characters (e.g. Cyrillic) still come from JetBrains Mono of the same weight.
- Along the way: Playwright **C16** (axis-constrained dragging, phase B), which recently failed even on its own, is fixed – the cause was in the test (handles of the previous cursor line, section 10).

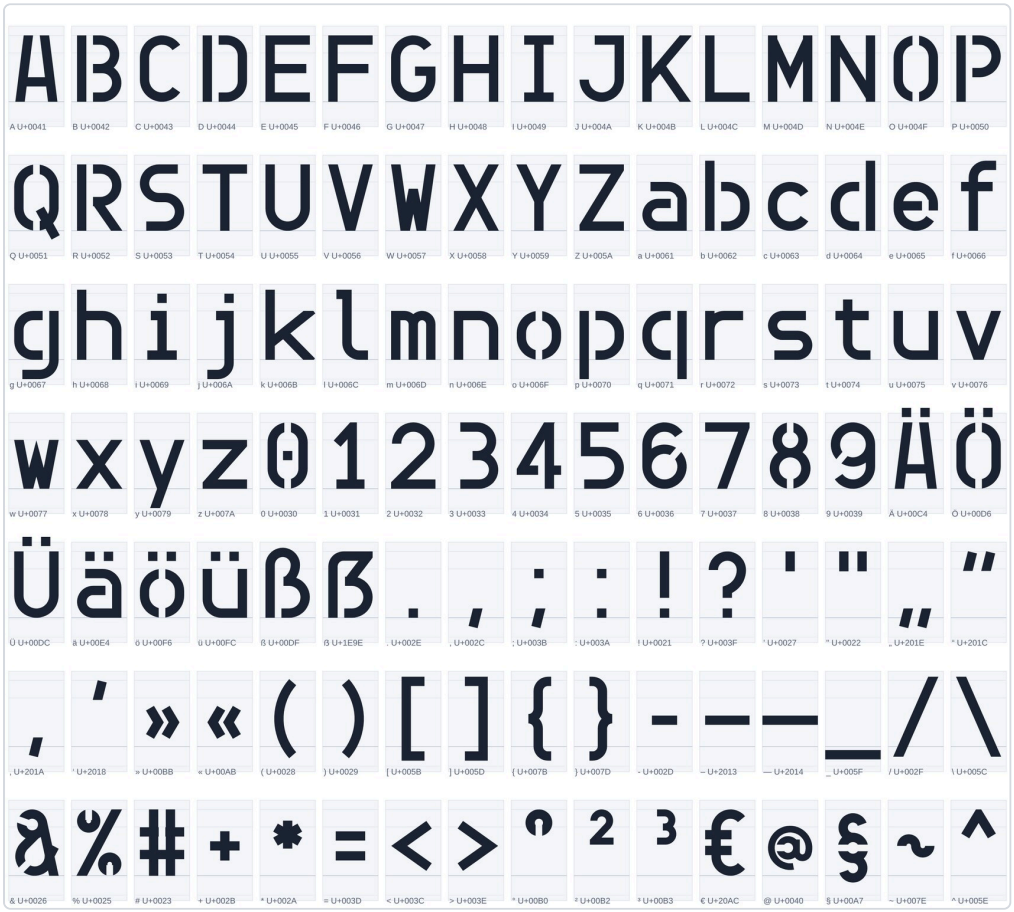
2 Decisions

Question	Decision
Form of the font	Rust data instead of a font file: per character a small drawing function over named positions (center lines left/right/middle, baseline, x-height, cap height, ascender, descender, depending on the stroke width), building blocks stadium, bowl, C arc, dots, commas. No TTF build, no license question; Wasm growth is code only (section 8).
Stroke	`region2::stroke` (from step 2) with flat ends and miter: right angles stay square, ends are straight cuts (stencil look). Free ends run to the outer line (baseline, cap height), joints end on the center line of the other stroke – so no notches occur. Diagonals (A, K, M, N, V, W, X, Y, Z, 1, 2, 4, 7, /) are extended beyond the line and trimmed at the baseline/cap line (straight feet, flat apexes).

Bridges	A bridge is a band of the bridge width across a stroke (length 1.3 half strokes to each side), subtracted from the glyph. Rules: stem separated from the bowl for B, D, P, R, b, d, g, p, q (bridges at the stem), round shapes at top and bottom (O, Q, 0, 8, o), open A (legs at the cap line apart by the bridge width), e and a at the crossbar or at the bowl next to the stem, 4 at the crossbar before the stem, 6/9 diagonally at the bowl, rings (°, %, Ø, @, &, §) on one side. ß/ß are open (gap ≥ bridge width to the stem).
Parameters	`stroke_width` and `bridge_width` in mm (for printing and milling, absolute dimensions are what matters: "bridge at least two nozzle widths"); if not given, proportional to `size`. Style Bold = default stroke width 0.2·`size`. The limits come from the test of the parameter space: above 0.2·`size` strokes of a, e, #, @ grow together and close counters; bridges below 0.03·`size` are closed again by heavy strokes.
Metrics	1000 units per em, cap height 720, x-height 530, ascender 760, descender −210, advance 600, side bearing 60 (glyph box 480 wide). Vertical text: advance 1000, glyph centered on the pen (as the shaper does for TrueType).
Default font and OpenSCAD	OpenSCAD does not know the font (without `font` it takes the fontconfig default font, usually Liberation Sans): a model without `font` looks different in OpenSCAD. The compatibility tests do not depend on it (section 9); anyone who needs the same dimensions as OpenSCAD specifies a bundled TrueType font. Unknown families → default font with a warning (previously JetBrains Mono).

3 Character chart

All 125 glyphs directly from the font (example program `cargo run -p cad-font --example stencil_chart`): cell = advance, lines baseline, x-height, cap height, descender. A glyph with a hole would be red.



cad stencil mono Regular (stroke 0.14·size, bridge 0.1·size)



cad stencil mono Bold (stroke 0.2·size, bridge 0.1·size)

Parts per glyph (Regular): 88× 1 part, 27× 2 parts, 8× 3 parts, 2× 4 parts; holes: Regular 0, Bold 0.

4 Parameters

The same eight characters with different stroke and bridge widths (fractions of `size`). `bridge_width = 0` turns the bridges off: then O, B, 0, 8 … have counters again (red = glyph with a hole, intended here).



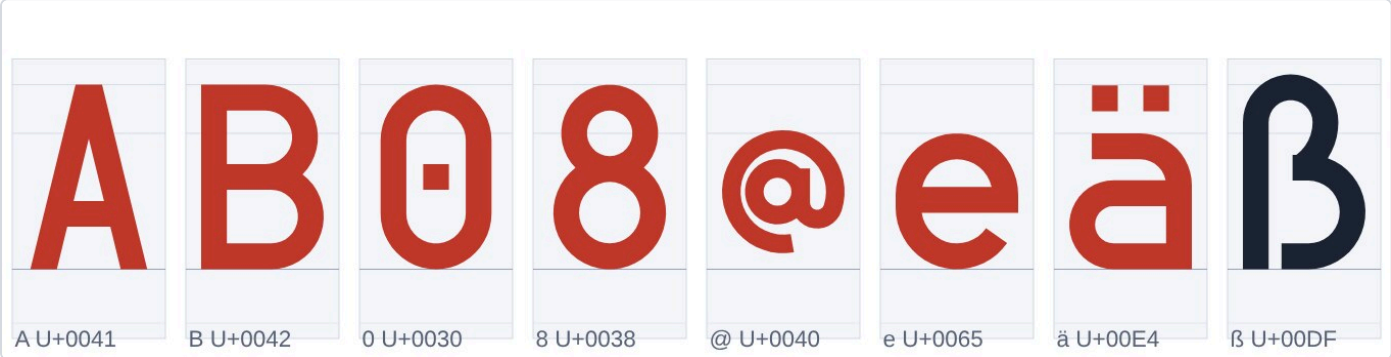
Stroke 0.06 · bridge 0.04



Default: stroke 0.14 · bridge 0.1



Stroke 0.2 · bridge 0.2 (limits)



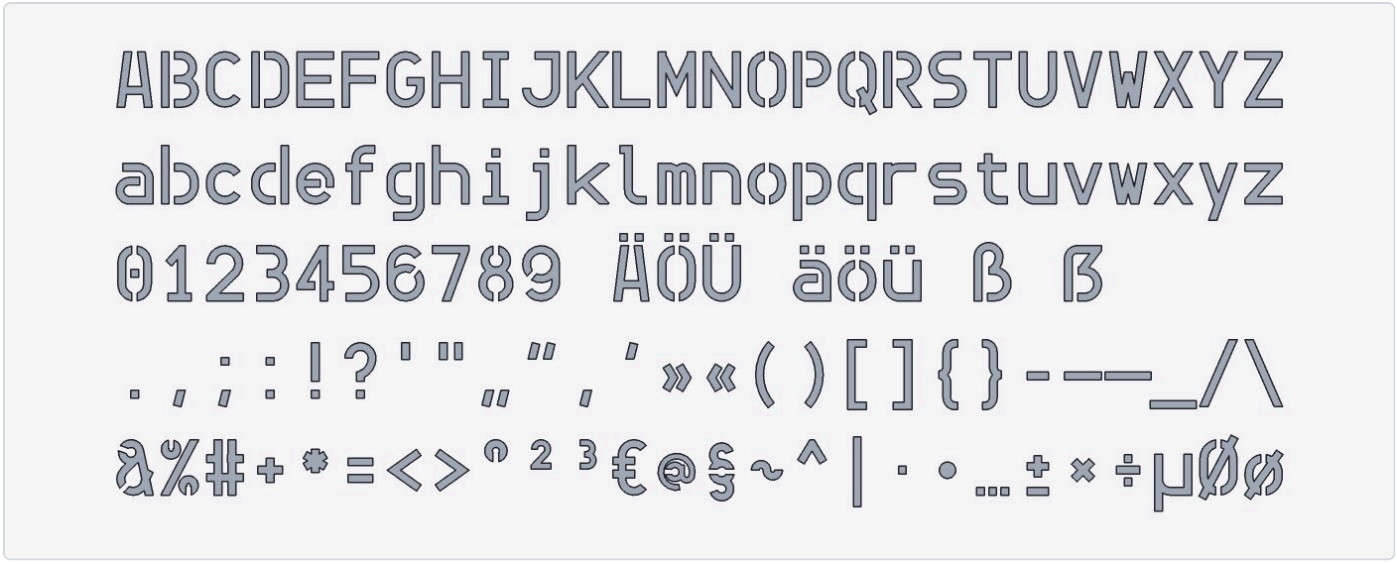
Stroke 0.14 · without bridges (bridge_width = 0)

Warnings for values outside the limits and for other fonts:

```
cases/warn.scad:1:37: Warnung (Auswertung): text: `stroke_width` = 3 mm bei size = 10: höchstens 2 mm
cases/warn.scad:1:55: Warnung (Auswertung): text: `bridge_width` = 0.1 mm bei size = 10: mindestens 0.3 mm
(oder 0)
cases/warn.scad:2:42: Warnung (Auswertung): text: `bridge_width` wirkt nur mit der Schrift „cad stencil mono“
cases/warn.scad:3:18: Warnung (Auswertung): Schrift „Comic Sans“ nicht vorhanden, verwendet wird die
Standardschrift cad stencil mono (mitgeliefert: cad stencil mono, JetBrains Mono, Inter, Allerta Stencil;
eigene Schrift: font = "datei.ttf")
cases/warn.scad: 0 Fehler, 4 Warnungen
```

5 Comparison with Allerta Stencil

Alphabet charts via the CLI (`text(…, size = 8)`, extruded 0.6 mm, from above): the stencil font has all characters itself; Allerta Stencil (proportional) lacks 20 of them (taken from JetBrains Mono, not stenciled, report 16).



cad stencil mono Regular



cad stencil mono Bold



Allerta Stencil – ß „ , { } _ # ° ² ³ § ~ ^ · • … ± µ Ø ø from JetBrains Mono

Same sign in both fonts



cad stencil mono (default) – volume 6281.4993 mm³



Allerta Stencil – volume 6294.8797 mm³

```
// sign with raised text (report 17): stencil
color("white") cube([100, 30, 2]);
color("black") translate([50, 17, 2]) linear_extrude(1) text("WERKSTATT 3", size = 8, halign = "center");
color("black") translate([50, 5, 2]) linear_extrude(1) text("ZUTRITT NUR MIT BRILLE", size = 4.2, halign =
"center");
```

Property	cad stencil mono	Allerta Stencil
Width	monospaced, 0.814·size per character (as JetBrains Mono)	proportional, with kerning
Curves	lines and circular arcs (arcs > 90° split in the kernel)	quadratic Bézier curves → one B-spline per flank
Required character set	complete, all stenciled	20 characters missing (fallback, without bridges)
Stroke, bridge	adjustable (`stroke_width`, `bridge_width`), Bold	fixed
License	source code of the project	SIL OFL 1.1, Reserved Font Name (unmodified only)

6 Example: stencil

`examples/spec/schablone.scad`: text cut out of a 1 mm thin plate. Without bridges the inner parts of A, 0, 8, B, 9 would fall out; with the stencil font they stay attached to the plate.

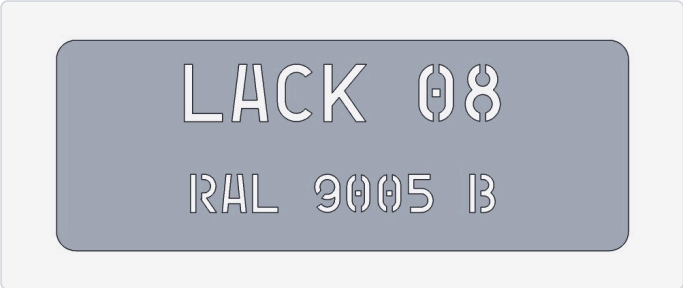
```
// Purpose: spray stencil with the default font "cad
stencil mono" (plan 17.4, report 17) – the
// text is cut out of a thin plate; the bridges of
the font hold the inner parts
// of O, A, B, 0, 8 … on the plate, nothing falls
out. Without `font` the stencil font applies;
// `stroke_width` (stroke width) and `bridge_width`
(bridge width) in mm, `bridge_width = 0` without
bridges.

breite = 120;
hoehe = 44;
dicke = 1;
r = 4;              // corner radius
groesse = 12;       // cap height
strich = 1.7;       // stroke width (default 0.14 ·
size)
steg = 1.2;         // bridge width (printing: at
least two nozzle widths)

difference() {
    linear_extrude(dicke) offset(r = r) offset(delta
= -r) square([breite, hoehe]);
    translate([breite / 2, 27, -1])
linear_extrude(dicke + 2)
        text("LACK 08", size = groesse, halign =
"center", stroke_width = strich, bridge_width =
steg);
    translate([breite / 2, 8, -1])
linear_extrude(dicke + 2)
        text("RAL 9005 B", size = 8, halign =
"center", bridge_width = steg);
}
```



rendered (iso)



top view

Printed: G-code from PrusaSlicer

A real print is not possible in this environment; instead: STL exported (`cad export`), sliced to G-code with PrusaSlicer 2.7 (default profile, layer 0.2 mm, nozzle 0.4 mm), and the top layer drawn from the G-code (`tools/report/report17-gcode.py`): every island of the plate is connected to the rest, the bridges (1.2 mm = 3 nozzle widths) are printed as separate paths.



top layer (z = 1 mm) from the G-code, 5 layers, estimated print time 53m 3s, filament 2058.04 mm

7 How a glyph is built

Example B: stem `vline(l, 0, cap)`; upper and lower bowl as a path from the stem (`bowl`: crossbar, quarter arcs, crossbar back); three bridges directly to the right of the stem (crossbars top, middle, bottom). `region2::stroke` turns the paths into bands, miter pieces and ends, `region2::combine` unites them and subtracts the bridges. Result: stem and "3"-shaped bowl as two closed contours of lines and arcs, without a hole. The positions depend on the stroke width (center line = outer line ∓ half stroke), so cap height and glyph box stay the same for every stroke width.

```
'B' => {
    g.vline(l, y0, cap);
    bowl(g, l, r - 0.06 * wd, t, m, (t - m) / 2.0);
    bowl(g, l, r, m, b, (m - b) / 2.0);
    for y in [t, m, b] {
        g.cut(l + h + bw / 2.0, y, 90.0);
    }
}
```

8 Measurements

Kernel time

Release build, `cad eval --json` without daemon, median of 3 runs; kernel time in ms (all kernel calls). The stencil font needs fewer kernel calls and less time than Allerta Stencil and JetBrains Mono: lines and circles instead of B-splines; its own computation time (regions, cached) is part of the evaluation (language 1.6 ms for the sign).

Model	cad stencil mono	Allerta Stencil	JetBrains Mono	Stencil / Allerta
"Hello World", extruded	130 (18 calls)	414 (26 calls)	417 (20 calls)	0.31
40 characters, extruded	962 (53 calls)	2,652 (95 calls)	2,126 (61 calls)	0.36
Sign: plate ∪ "ZUTRITT NUR MIT BRILLE" (size 4.2)	1,791 (33 calls)	3,419 (45 calls)	2,238 (34 calls)	0.52

Browser (Chromium, SMLib as Wasm, Playwright K03): sign 2,196 ms versus 3,208 ms with Allerta Stencil; "Hello World" 206 versus 463 ms. Stencil example native: 2,609 ms.

Wasm module

	raw	gzip -9
`cad_wasm_bg.wasm` before (commit ed85270)	4,120,239 B	1,757,290 B
with stencil font	4,162,774 B	1,772,667 B
growth	42,535 B	15,377 B

Same build (release, wasm-opt -O3) for both states. `region2` was already in the module (offset, stroke); only the drawing functions are added.

9 OpenSCAD compatibility

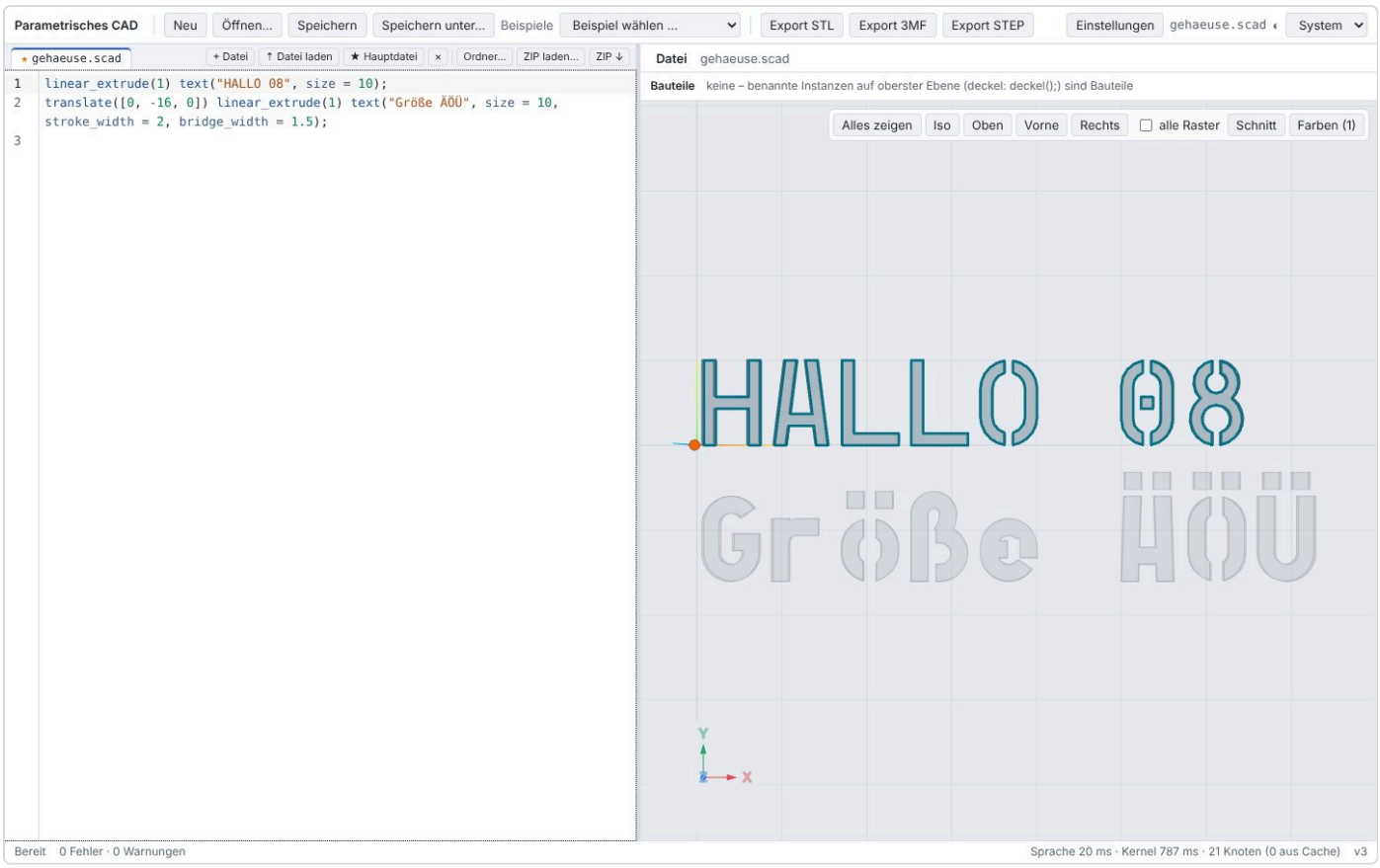
Do the compatibility tests depend on the default font? No: the 36 comparison models (`examples/openscad-compat`, `tests/openscad-compat/projects`) contain no `text()` (grep finds no file), so `compare.py` and the stage 1 tests (`crates/compat-tests`) are unaffected; the text comparison with OpenSCAD from report 16 (`hull_and_area_like_openscad`, `report16-openscad.py`) names every font explicitly. Run `compare.py --eval`: 36/36 files match. Intentional deviation: `text("…")` without `font` gives the stencil font here and the fontconfig default in OpenSCAD – for identical dimensions, specify a font (plan 17.11, skill).

10 UI

There is no font selection in the UI (fonts are specified in the code; custom ones via "↟ Load file" (German UI: „↟ Datei laden“)), so no change there. New scenarios in `web/e2e/schablone.spec.ts`:

K01 Default font in the browser

The Wasm engine uses the same font as the CLI: cap height = size, advance as JetBrains Mono, umlauts without fallback.



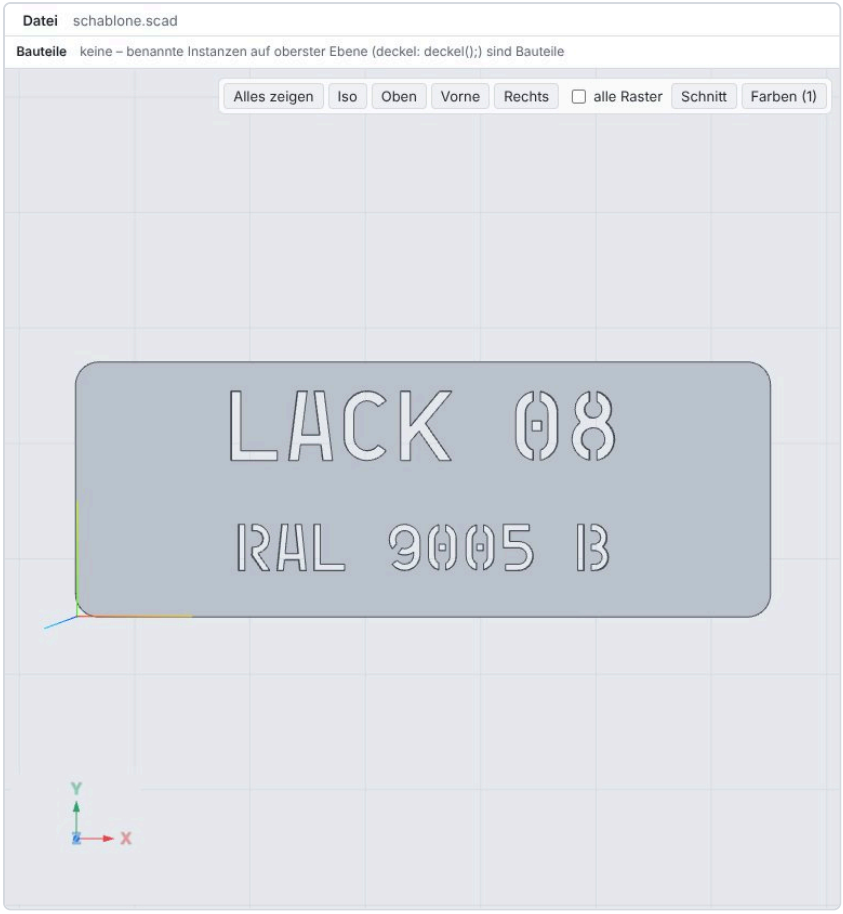
text() without font: cad stencil mono, second line with stroke_width = 2, bridge_width = 1.5

```
linear_extrude(1) text("HALLO 08", size = 10);
translate([0, -16, 0]) linear_extrude(1) text("Größe ÄÖÜ", size = 10, stroke_width = 2, bridge_width = 1.5);
```

- no messages
- cap height 10 mm, height 1 mm
- width 7–9 cells
- limits and unknown font: warnings

K02 Stencil example

Plate minus text: the inner parts of A, 0, 8, B, 9 are attached to the plate via the bridges.



examples/spec/schablone.scad in the browser (top view)

- no errors/warnings
- bounding box 120 × 44 × 1 mm

C16 axis-constrained dragging (report 4) – cause and fix

Since report 15, C16 ran into the test timeout depending on timing, recently also on its own. Cause: after `clickLine(2, 3)` the viewer rebuilds the handles only in the next animation frame; until then `__cad.handles()` returns the handles of the previous cursor line. That line holds `cube([60, 30, 1])`, whose vector is also a point handle – `waitHandles("… kind === 'point'")` was satisfied immediately, the test took the first handle (the plate corner at x = 1118 px instead of the translate point at 770 px), the drag did not start (`drag() = null`), the mouse rotated the camera, and waiting for "Achse x" ("axis x") ran until the timeout. The axis selection itself was correct in all runs. Fix in the test: `pointHandleOn(page, zeile)` waits for the point handle **of the line** and takes exactly that one (likewise C20, same gap). No test skipped.

```
before (pnpm exec playwright test e2e/ui-c.spec.ts -g C16, several runs with --repeat-each): 10 of 16 passed, 6
× test timeout (waitForFunction drag()?.axis === 0)
  failed run: handle at start of wait {"line":1,"value":[60,30,1],"screen":{"x":1117.7,"y":527.9}} (cube of
line 1), drag() = null
after (pointHandleOn): C16 16 of 16, C20 8 of 8 passed (approx. 25 s and 10 s each)
```

11 Test runs

Run	Result
`cargo test --workspace --no-fail-fast`	529 passed, 0 failed
`cargo clippy --workspace --all-targets -- -D warnings` · `cargo fmt --check`	clean · clean
`pnpm typecheck` · `pnpm test` (Vitest)	clean · 221 passed
`pnpm e2e` (Playwright, Chromium)	106 passed, 0 failed

`tests/openscad-compat/compare.py --eval`	36/36 files match

New or changed Rust tests (13):

`every_glyph_profile_has_no_holes` **ok** · `extruded_text_volume` **ok** · `sign_text_kernel_time_like_allerta` **ok** · `text_stencil_parameters` **ok** · `text_profiles_and_fonts` **ok** · `tests::font_names` **ok** · `tests::stencil_areas` **ok** · `tests::required_chars_in_every_font` **ok** · `tests::stencil_layout_features` **ok** · `tests::stencil_metrics_and_monospaced_advance` **ok** · `tests::stencil_without_bridges_has_counters` **ok** · `tests::stencil_glyphs_have_no_holes` **ok** · `spec_models_stage1` **ok**

cad-font: coverage (`required_chars_in_every_font` with both styles), no holes/slivers/overlaps for 8 parameter combinations, closed counters without bridges, metrics (cap height = size, advance = JetBrains Mono, every glyph in its cell), areas (H exact, o against formula, alphabet against JetBrains Mono, stroke/bridge/Bold/size), alignment, directions, fallback, font names. cad-eval: default font, parameters, warnings. cad-engine: every glyph as a profile without a hole (Regular, Bold), volume = area × height, kernel time of the sign against Allerta Stencil. compat-tests: example `schablone.scad`.

12 Limitations

- Design deliberately simple: one stroke width per glyph (only @ with 0.72 times the stroke, ² ³ with 0.75 times), no optical corrections, no kerning pairs (monospaced). W, w, m, @, * become very dense in Bold.
- Parameter limits from the test (stroke ≤ 0.2·size, bridge 0 or 0.03…0.2·size); values in between checked at 8 combinations, not exhaustively. On arcs, a bridge is cut radially straight; its width applies on the center line.
- The font does not compose combining characters (e.g. "a" + U+0308): the character comes from JetBrains Mono. Precomposed umlauts are included.
- Characters outside the scope (Cyrillic, Greek, arrows …) come from JetBrains Mono – without bridges, so they can have holes.
- Glyphs of different characters touch only with `spacing` < 1 or with the continuous strokes –, _ (across the whole cell as in monospace fonts); then counters can form between glyphs.
- Models without `font` look different in OpenSCAD (fontconfig default there).